INCH-POUND
MIL-DTL-24643/34G
27 November 2023
SUPERSEDING
MIL-DTL-24643/34F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPE LSMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First	-	Copper conductor, tin-coated, 16 AWG, class B stranding (7 by 0.0192 inches), in accordance with ASTM B8 or ASTM B286, 16 AWG with 19 strands.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation, white. Minimum average wall thickness of 0.013 inch. Standard identification code applied by method 1.
Fourth	-	The specified number of singles (see table I) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.
Fifth	-	An optional binder.
Sixth	-	Braided shield of tin-coated copper.
Seventh	-	An optional binder.
Eighth	-	Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.075 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

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TABLE I. Details.

Military part no. M24643/34	Type and size	Number of conductors	Overall diameter (inch)	
			min.	max.
-01UO	LSMS-37	37	0.740	0.800

NOTE:
1. Weight of completed cable shall not exceed 0.060 lb/ft.

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	5.1
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Conductor to overall shield	1,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	2,000
Conductor to overall shield	500
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150

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Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60
Group D:	
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2

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Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSMS-37	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-050)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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